

ASSIGNMENT NO. 2

✓1 IDL File (Calculator.idl)

□ IDL (Interface Definition Language) defines remote methods.

```
module CalculatorApp {  
  
    interface Calculator {  
  
        float add(in float a, in float b);  
        float subtract(in float a, in float b);  
        float multiply(in float a, in float b);  
        float divide(in float a, in float b);  
  
    };  
};
```

CalculatorImpl.java

```
import CalculatorApp.*;  
  
class CalculatorImpl extends CalculatorPOA {  
  
    public float add(float a, float b) {  
        return a + b;  
    }  
  
    public float subtract(float a, float b) {  
        return a - b;  
    }  
  
    public float multiply(float a, float b) {  
        return a * b;  
    }  
  
    public float divide(float a, float b) {  
        if (b == 0)  
            return 0;  
        return a / b;  
    }  
}
```

CalculatorServer.java

```
import CalculatorApp.*;  
import org.omg.CosNaming.*;  
import org.omg.CosNaming.NamingContextPackage.*;  
import org.omg.CORBA.*;  
import org.omg.PortableServer.*;  
  
class CalculatorServer {  
    public static void main(String args[]) {  
        try {  
  
            ORB orb = ORB.init(args, null);  
  
            POA rootpoa = POAHelper.narrow(  
                orb.resolve_initial_references("RootPOA"));  
            rootpoa.the_POAManager().activate();  
  
            CalculatorImpl calcImpl = new CalculatorImpl();
```

```

        org.omg.CORBA.Object ref =
rootpoa.servant_to_reference(calcImpl);
        Calculator href = CalculatorHelper.narrow(ref);

        org.omg.CORBA.Object objRef =
            orb.resolve_initial_references("NameService");

        NamingContextExt ncRef =
            NamingContextExtHelper.narrow(objRef);

        NameComponent path[] = ncRef.to_name("Calculator");
        ncRef.rebind(path, href);

        System.out.println("Calculator Server ready...");

        orb.run();

    } catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

✓ 5 Client Program

CalculatorClient.java

```

import CalculatorApp.*;
import org.omg.CosNaming.*;
import org.omg.CORBA.*;
import java.util.Scanner;

class CalculatorClient {
    public static void main(String args[]) {
        try {

            ORB orb = ORB.init(args, null);

            org.omg.CORBA.Object objRef =
                orb.resolve_initial_references("NameService");

            NamingContextExt ncRef =
                NamingContextExtHelper.narrow(objRef);

            Calculator calc =
CalculatorHelper.narrow(ncRef.resolve_str("Calculator"));

            Scanner sc = new Scanner(System.in);

            System.out.println("Enter two numbers:");
            float a = sc.nextFloat();
            float b = sc.nextFloat();

            System.out.println("Addition: " + calc.add(a, b));
            System.out.println("Subtraction: " + calc.subtract(a, b));
            System.out.println("Multiplication: " + calc.multiply(a, b));
            System.out.println("Division: " + calc.divide(a, b));

        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}

```

}

✓Sample Output

